# Example Questions for Practice

**ECE373**
**Dr. Marefat**

Multiple Choice (40 points, 5 each)

1. The default constructor that Java provides

A. sets all the class's numeric fields to 0, boolean values to false, and reference variables to null.
B. sets all the class's numeric fields to 0, boolean values to true, and reference variables to empty strings.
C. sets all the class's numeric fields to 0, boolean values to true, and reference variables to null.
D. sets all the class's numeric fields to 0, boolean values to false, and reference variables to empty strings.

2. If the following were all declared in the same class, *Rect*, which would be considered the no-arg constructor?

(1) public Rect(int len, int width){...}
(2) public Rect(double len, double width){...}
(3) public Rect(){...}
(4) private Rect(){...}

A. (1)
B. (2)
C. (3)
D. (3) and (4)

3. If object1 and object2 are references to objects of the same class, to make object2 a copy of object1, the best approach is:

A. Use the default constructor to create object2 with object1 data members
B. Assign object1 to object2 like this: object2 = object1;
C. Write a copy method that will make a field by field copy of object1 data members into object2 data members
D. Use the copy method that is inherited from the class Object

4. Which of the following is indicative of an aggregation relationship between two classes:

A. Is-A relationship
B. One class is an inner class of the other
C. Has-a relationship
D. One class is a subclass of the other

5. Assume the class BankAccount has been created, and the following statement correctly calls a constructor to create an instance of the class:

```
BankAccount account = new BankAccount(5000.0);
```

When the following statement is executed:

```
System.out.println(account);
```

A. A runtime error will occur
B. It will print 5000.0
C. The toString method will be called
D. It will print binary numbers corresponding to the binary representation of the object account

6. A method's signature consists of

A. the method name and the parameter list and their types.
B. the return type, the method name, and the parameter list and their types.
C. the method name.
D. the return type and the method name.

7. Which of the following is true about the static methods of a class:

A. They must be declared public.
B. They can only refer to the static members of their class unless they have a class instance reference.
C. They can only refer to non-static members of the class unless they have a class instance reference.
D. An instance of the class should be created to call them.

8. Which of the following is true if the reference *this* is used in a constructor to call another constructor from the same class:

A. We cannot use *this* in a constructor
B. It must refers to a parameter in a method
C. A compiler error will occur if it is before any other line in the method
D. If it is used to call another constructor, it must be the first line

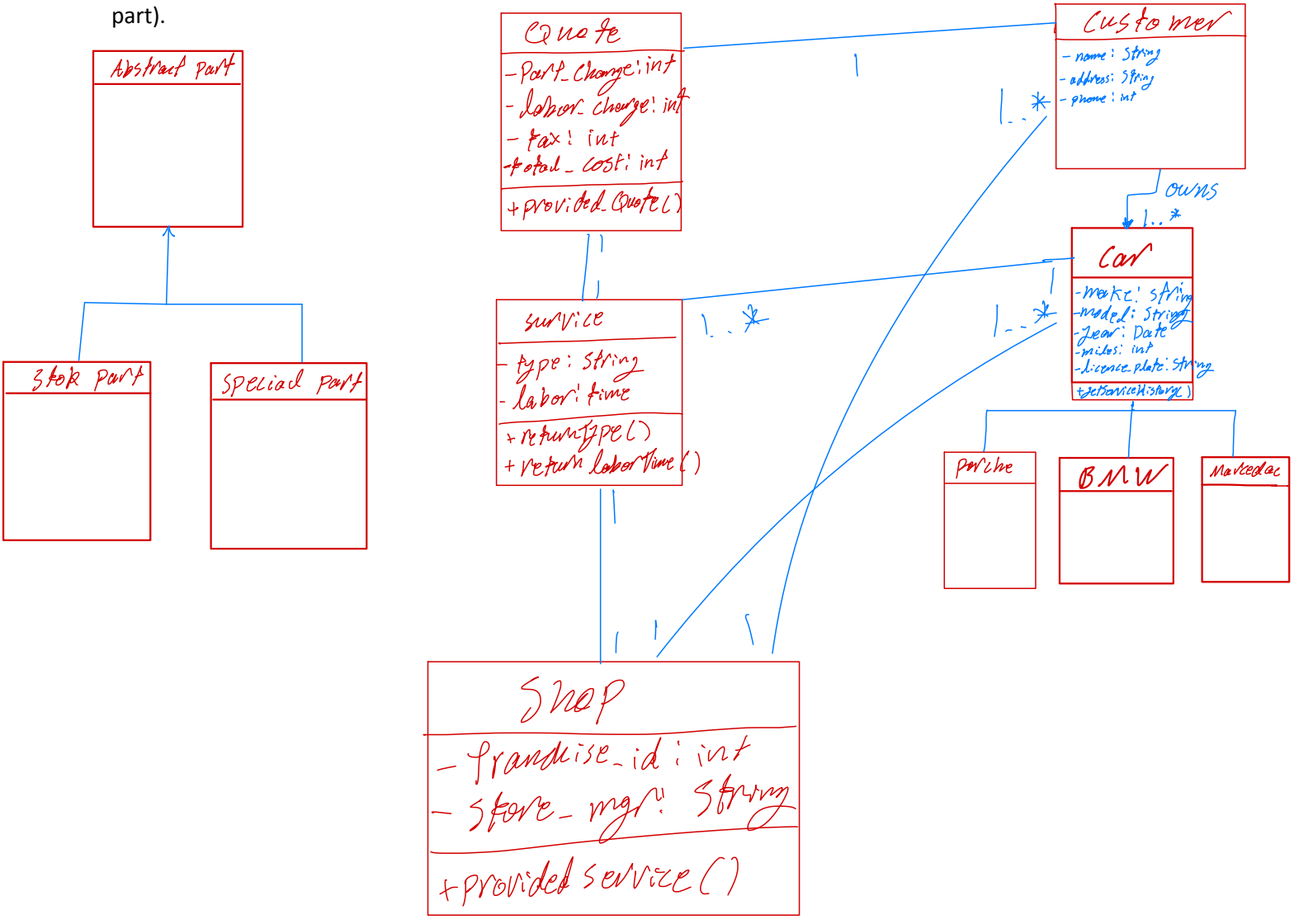
Analysis Problem

**1. (50 points, 25 each for parts A and B)**

Joe's automotive franchise services foreign cars, and specializes in cars made by BMW, Porsche, and Mercedes. There are shops at different locations, each with its own franchise id, and its own store manager. When a customer brings in a car to the shop, the manager gets the customer's name, address, and phone no. The manager then determines the make, model, and year of the car and records how many miles it has and its license plate. Then he gives the customer a service quote. The quote includes estimated part charges, the estimated labor charges, the tax, and the total estimated cost. Each type of repair or service has its own expected labor time which is used to determine labor charges by multiplying with the hourly rate for a mechanic. The estimated part charge is calculated based on the sum of the prices for the parts involved in the given service. If a car has been to any of the franchise shops before, the manager should be able to look at what services/repairs (and the service/repair's details) have been done in the car's service history.

A. Analyze the problem description and present a Domain class model for developing the software for Joe's automotive franchise in UML.

B. Show declarations that implement the model developed in part A in Java. You need to include/declare getter and setter methods only if they are being used by some other method in the model (for example the price for a part).



**2. Short Answer Problem (10 points)**

**Look at the following classes:**

```
public class SubClass1 extends SuperClass1
{
  public SubClass1()
  {
    System.out.println("This is the subclass " + "constructor.");
  }
}

public class SuperClass1
{
  public SuperClass1()
  {
    System.out.println("This is the superclass " + "constructor.");
  }
}
```

**What does the following program display (assume the program compiles):**

```
public class ConstructorDemo
{
  public static void main(String[] args)
  {
    SubClass1 obj = new SubClass1();
  }
}
```

This is the superclass constructor.

This is the subclass constructor.